ILPO-JUHANI VÄÄTÄINEN

Independent measurement research — recomputing published numbers with their uncertainty attached
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PROFILE

I take numbers that have already been published and recompute them with their uncertainty attached, then state what the data does and does not license. The work is narrow on purpose: given an organisation's own published figures, I establish the smallest difference its evidence could have detected, and whether the differences it reports clear that bar.
This matters most where a number crosses a threshold or drives a decision — a classification boundary, a results-framework indicator, a declared accuracy level, a league table. There an unstated uncertainty is the difference between a defensible finding and one that is withdrawn later. Every figure I report is recomputed through a second, independent code path before I believe it, and the code is published with the finding.

SELECTED WORK

Coverage versus reported regional differences — national school health survey, 2026
A publisher reports coverage by region (63–78% at one school level, 22–49% at another) and justifies comparability by the large number of respondents and the confidence intervals. The interval does not address the coverage difference. Between the two extreme regions the difference arising from non-response alone is 0.15 x however much non-respondents differ from respondents; the published interval is computed as though that were zero.

Classification thresholds versus measurement tolerance — surface water status, 2026
The classification criteria (182 pages) contain no occurrence of measurement uncertainty, analytical uncertainty or laboratory, and bundle different determination methods as comparable. For one coastal type the entire "good" class for total phosphorus spans 11–13 ug/l — 15% of its upper boundary — while the same institute's proficiency testing accepts 5–25% deviation. Two accredited laboratories can both pass and place the same water on opposite sides of the line.

Declared accuracy without its evidential basis — AI accuracy standardisation, 2026
Written technical input to the European Commission's AI Office on standardisation request M/593: a declared false-acceptance rate of 1e-6 supported by an error-free evaluation over 1,000,000 comparisons cannot be carried by that evaluation, since with zero errors in n trials the 95% upper bound is 3/n — three times above the declared figure. Clause text supplied to the national

mirror committee and the technical committee secretariat.

A benchmark headline that is the label's own algebra — logistics, 2026

A widely used late-delivery benchmark reports 97% accuracy. Reproduction shows the figure is recoverable from the label's own construction rather than predicted, with the leaked-versus-clean comparison and the honest number underneath.

PUBLICATIONS AND ARCHIVED CODE

All deposited with the code that produced the numbers. Concept DOIs resolve to current versions.

How much of a published statistic belongs to the choice upstream of it: a protocol
10.5281/zenodo.22162050

The Benchmark Number Is Not the Deployment Number: rank collapse and base-rate collapse
10.5281/zenodo.22109472

How the choice of sign inventory moves the statistics of an undeciphered script
10.5281/zenodo.22057706

A Pre-Registered Configuration Test for the Göbekli Tepe Pillar 43 "Date Stamp"
10.5281/zenodo.22139090

Eval Integrity: independent audits of the measurement behind published AI evaluation numbers
10.5281/zenodo.22343058

The 97% That Predicts the Past: leakage in a logistics late-delivery benchmark
10.5281/zenodo.22342749

Leaderboard measurement toolkit: how much of a ranking survives its own sampling error
10.5281/zenodo.22342751

ControlBattery: machine-checked statistics for a control battery (Lean 4 / mathlib)
10.5281/zenodo.22342753

The constant that caught my CV lying: a preregistered prediction and the miss that refuted it
10.5281/zenodo.22342755

Book: Measured, Not Believed — How AI Benchmarks and Trading Backtests Overpromise

OPEN-SOURCE CONTRIBUTIONS

Merged upstream, owned end to end including tests and review:

ffn (financial analysis library) x2 · Uber causalml x1 · Hummingbot (trading framework) x3

SOFTWARE AND SECURITY

Independent smart-contract security research across DeFi protocol types (vaults, lending, AMMs, bridges, staking) with findings submitted to live bug-bounty programmes; Foundry proof-of-concept, invariant and fork testing, Slither and Halmos. Representative finding: a share-inflation bug in a vault where shares were minted against the contract's own token balance, validated end to end and adversarially re-tested to quantify value-at-risk before reporting.

Delta-neutral vault deployed to Solana mainnet (Honorable Mention, 2026 hackathon); mobile application published on Google Play.

SKILLS

Statistics: uncertainty propagation, power and detectable difference, non-response and coverage, multiple comparisons, calibration and base rates, preregistration

Tools: Python (numpy, scipy, pandas), Lean 4 / mathlib, R basics, SQL, Foundry, Slither, Halmos

Practice: reproducible pipelines, independent re-derivation of every reported figure, published code

Languages: Finnish (native), English (fluent, written and spoken)

EDUCATION

Datanomi — Vocational Qualification in Information and Communications Technology

Keuda, Kerava, Finland · 2009–2011

STATED LIMITS

Given so that a reader does not have to ask. No doctorate. No prior paid client assignment in evaluation — the work above is self-initiated and published rather than commissioned. Fieldwork, interviewing and facilitation are not strengths; data, methods and written analysis are. I work in writing rather than by phone. I build software by directing AI tools and say so; the numbers are checked by re-derivation, not by trust.